Fiscal Year:	FY 2008	**Task Last Updated:**	FY 12/16/2008
PI Name:	Bloomfield, Susan A. Ph.D.		
Project Title:	Increasing the Efficiency of Exercise Countermeasures for Bone Loss		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-4375	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
Start Date:	04/01/2004	**End Date:**	04/30/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	8	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 4/30/2008 per K. Major/NSBRI (3/08)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Baldwin, Kenneth (University of California, Irvine) Hogan, Harry (Texas A&M University)		
Grant/Contract No.:	NCC 9-58-BL00404		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

The over-arching goal of this project was to test the hypothesis that astronaut exercise time devoted to mitigation of bone loss can be reduced if combined with an anti-resorptive pharmacological agent. Skeletal muscle mass, function, and anabolic markers were also assessed to assure that any reductions in exercise time did not sacrifice benefits to muscle. The original aims of the project were to 1) test two different modes of resistance training (electrically stimulated muscle contractions versus voluntary resistance exercise) for their effects on mitigating bone and muscle decrements during 28 days of hindlimb unloading (HU) in adult rats; 2) to test different doses of alendronate during 28 days HU for maximal efficacy in suppressing bone resorption, and then 3) to combine alendronate with that training paradigm judged most successful to test for additive or synergistic effects.
The first year of this project was largely devoted to the adaptation of a flywheel device for voluntary resistance training in HU rats and to modifications of our existing muscle stimulation system, which allows for precisely controlled (but involuntary, therefore less physiological) muscle contraction regimens of the rat's lower leg. This latter system became the training paradigm of choice for the remainder of the project. In Year 3, a revision of the simulated resistance training protocol was completed in hopes of maintaining the positive impact on bone mass and strength and providing some mitigation of loss of skeletal muscle function and mass. The primary changes in the training paradigm were 1) decreasing the intensity of muscle contractions; 2) changing the range of motion allowed at the ankle during lower leg contractions, to achieve a greater stretch on the posterior crural muscle group; and 3) adding 1 second of isometric muscle contraction to the 1 second of eccentric contraction. This training regimen proved even more effective for mitigating negative bone changes, now impacting positively on bone geometry and mineral content at the mid-shaft tibia as well as at the proximal site. To our knowledge, this is the first exercise countermeasure regimen to demonstrate not just prevention of loss but an absolute gain in bone mass during simulated microgravity. However, this same sRT produced only minor mitigation of decrements in muscle strength and mass, nor were assays results for anabolic markers in gastrocnemius muscle encouraging. Parallel strain gage studies during simulated resistance training are providing unique information about the nature of the strain stimulus experienced by bone during isometric and eccentric muscle contractions; these data are being developed into a separate manuscript.

In our final year, we completed a record number of experiments. A dose finding experiment with 3 doses of alendronate was completed in order to define the minimally efficacious dose of alendronate, which turned out to be 0.01 or 0.03 mg/kg (delivered 3 days/week), depending on which variable of bone mass or geometry was examined. A small n trial testing the efficacy of sRT performed at a lower intensity (75 vs 100% peak isometric torque) verified that this reduced intensity produced more promising results for skeletal muscle while retaining its prevention of disuse-induced bone loss.

Our final animal experiments tested the separate and combined effects of reduced-volume sRT [reduced intensity along with a reduced frequency (2 days/week)] with the lowest dose of alendronate exhibiting any positive mitigation of bone loss (0.01 mg/kg BW). The pQCT data indicate that sRT effectively mitigated the losses of bone mineral content/density, especially in the cancellous compartment of the proximal tibia which is that bone compartment most affected by the unloading of tail suspension. In fact, for some variables, HU rats on sRT alone experienced a gain in bone during the unloading period. This was remarkable, given the reduced intensity and frequency of training used in this last protocol. Salutary effects were also observed at the mid-shaft tibia with sRT. New to this particular sRT stimulation protocol, we demonstrated effective maintenance of lower leg muscle strength: peak isometric torque of the lower leg muscles was effectively maintained in rats subjected to the sRT vs. the 10% decrease observed in rats on HU alone.

Alendronate treatments during 28 days HU were effective in mitigating bone loss only at the proximal tibia but appeared to have no effect on preventing the commonly observed disuse-induced suppression of periosteal expansion of mid-shaft tibia. Combining alendronate with sRT did not result in additional bone benefits beyond those observed with the training alone. The unexpected success of this reduced volume-sRT in preventing HU-induced bone loss, and on occasion stimulating absolute gains in bone mass, prevents us from answering one of our central hypotheses: that a marginally effective exercise countermeasure, combined with anti-resorptive effects of alendronate, would effectively mitigate bone loss. Hence, our preliminary conclusion is that the mechanical (and/or metabolic) stimulation of bone with high-force muscle contractions of an isometric/eccentric nature, even if performed only twice per week, appears to be an effective countermeasure for bone loss with unloading and can maintain ankle plantarflexor muscle strength if not muscle mass. This provides an important proof of concept: if resistance training includes muscle contractions at high enough intensities, with isometric and eccentric components, significant mitigation or prevention of bone loss and of reduced muscle strength is achievable. We expect that assays in progress (skeletal muscle anabolic markers, detailed histomorphometric studies of cancellous bone) will not modify this conclusion, but will provide important tissue- or cell-level mechanisms for these effects.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Our multiple trials testing the bone and muscle response to resistance training applied as a countermeasure during a period of simulated microgravity provide an important proof of concept: if resistance training performed during a period of prolonged non-weight bearing includes isometric and eccentric muscle contractions at high enough intensities, significant mitigation of bone loss and reduced muscle strength are achievable. Remarkably, we demonstrated that 40 seconds worth of muscle contraction, performed only twice per week, effectively prevented bone loss at the proximal tibia (a disuse-sensitive site) and maintained peak strength of the lower limb muscles. We also confirmed that the anti-resorptive agent alendronate, given at 1-3 times the dose level effective against bone loss due to estrogen deficiency, can effectively mitigate losses of bone during disuse. However, alendronate's salutary effect appears to be limited to metaphyseal bone (e.g. proximal tibia), whereas our resistance training paradigm had beneficial effects at that site and on mid-shaft bone as well. One clear implication of these findings for the public and for health care professionals is that small volumes of isometric plus eccentric-based strength training may be effective in minimizing loss of bone mass and skeletal integrity during prolonged periods of disuse (e.g., prolonged bed rest). This has important implications for rehabilitation specialists, physical therapists, and clinicians treating bedridden patients.

Task Progress:	1. Completion of dose-finding experiments for alendronate efficacy in preventing hindlimb-unloading (HU) bone loss: By late summer 2007 we completed experiments testing the efficacy of 3 different doses of our anti-resorptive pharmacological agent, alendronate, in mitigating bone loss during 28 days HU. Twenty-eight animals completed 28 days of HU: 6 received vehicle injections, while others received low (n=7), medium (n=8) or high (0.10 mg/kg, n=7) doses of alendronate. In vivo CT scans were performed on Days 0 and 28 to provide quick feedback to enable us to proceed to the final experiments. 2. Completion of trials to define best muscle stimulation protocol to test “reduced volume” training in final experiments: In the fall of 2007 we completed trials of 28-d HU on 15 animals to testing efficacy of a reduced training volume (fewer days/week at the same or reduced intensity vs previous experiments) in preventing loss of muscle mass and strength while retaining the beneficial bone effects previously observed. 3. Completion of final experiments to test combination treatment (anti-resorptive agent alendronate) and reduced volume of simulated resistance training: Starting late fall of 2007 and running through March 2008, we ran the labor-intensive experiments on 55 animals to test this project’s main hypothesis: resistance training time can reduced if combined with alendronate treatments during a period of simulated microgravity. A few additional animals were used to characterize the magnitude of the loading stimulus to bone by quantifying bone surface strain with rosette strain gages (MS thesis project of Mr. Scott Bouse). 4. Muscle samples shipped to collaborator Ken Baldwin’s laboratory: In May 2008 all accumulated gastrocnemius muscle samples from the experiments described in items 2 and 3 were shipped to Dr. Baldwin’s laboratory for analyses of early changes in markers for anabolic changes. As of this writing, those assays have been completed. 5. In progress: One manuscript is nearly completed as of this writing summarizing earlier trials testing simulated resistance training during 28 days HU, comparing two different stimulation paradigms (Swift JM et al.). Another is in progress defining results from an earlier muscle stimulation paradigm, combined with intriguing strain gage data collected during the period of hindlimb unloading (Hogan HA et al.). Once assays listed below are completed, at least two more manuscripts will be underway focusing on the impact of sRT on skeletal muscle and on details of cancellous bone microarchitecture/bone formation rate. ; a. Analysis of data from muscle anabolic marker assays (Dr. Baldwin’s laboratory). b. Histomorphometric assessments of proximal tibia cancellous bone. c. Analyses of rosette strain gage data gained during muscle stimulation training sessions of final experiments, likely useful adjunct data for our final manuscript describing efficacy of combining simulated resistance training and alendronate.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Sumner LR, Nilsson MI, Swift JM, Jeffery JM, Walthall CJ, Stallone JL, Bloomfield SA, Hogan HA. "Electrically stimulated eccentric muscle contraction benefits bone in the tibia of hindlimb unloaded rats at both the mid-diaphysis and proximal metaphysis." American Society for Bone and Mineral Research 29th Annual Meeting, Honolulu, HI, Sept. 16-19, 2007. J Bone Miner Res 2007 Sep;22(Suppl 1):S488. , Sep-2007
Abstracts for Journals and Proceedings	Swift JM, Ali JT, Stallone JL, Hogan HA, Bloomfield SA. "Beta-adrenergic receptor agonist administration during hindlimb unloading attenuates losses in bone geometry, structure, and mechanical properties." American Society for Bone and Mineral Research 29th Annual Meeting, Honolulu, HI, Sept. 16-19, 2007. J Bone Miner Res 2007 Sep;22(Suppl 1):S134. , Sep-2007
Articles in Peer-reviewed Journals	Swift JM, Swift SN, Allen MR, Bloomfield SA. "Beta-1 adrenergic agonist treatment mitigates negative changes in cancellous bone microarchitecture and inhibits osteocyte apoptosis during disuse." PLoS One. 2014 Sep 11;9(9):e106904. eCollection 2014. http://dx.doi.org/10.1371/journal.pone.0106904 ; PubMed PMID: 25211027; PubMed Central PMCID: PMC4161377 , Sep-2014
Articles in Peer-reviewed Journals	Mantri AV, Allaway HCM, Brezicha JE, Hogan HA, Bloomfield SA. "Oral estradiol impact on mitigating unloading-induced bone loss in ovary-intact rats." Aerosp Med Hum Perform. 2021 Feb 1;92(2):65-74. https://doi.org/10.3357/AMHP.5668.2021 ; PMID: 33468286 , Feb-2021
Articles in Peer-reviewed Journals	Baek K, Barlow AA, Allen MR, Bloomfield SA. "Food restriction and simulated microgravity: effects on bone and serum leptin." J Appl Physiol. 2008 Apr;104(4):1086-93. PMID: 18276897 , Apr-2008
Articles in Peer-reviewed Journals	Baek K, Bloomfield SA. "Beta-adrenergic blockade and leptin replacement effectively mitigate disuse bone loss." J Bone Miner Res. 2009 May;24(5):792-9. (Note reported previously in Dec 2008 as 2008 Dec 29. [Epub ahead of print]) https://doi.org/10.1359/jbmr.081241 ; PMID: 19113907 , May-2009
Awards	Baek K. "Distinguished Graduate Student Award for Excellence in Doctoral Research, Texas A&M University, March 2008." Mar-2008
Awards	Bloomfield S. "Invited lecturer for Texas Regional Chapter, American College of Sports Medicine Spring Lecture Tour, March 2008." Mar-2008
Awards	Swift JM, Bloomfield SA. "Plenary poster at annual meeting of the American Society of Bone & Mineral Research Annual Meeting, Honolulu, Hawaii, September 2007." Sep-2007
Books/Book Chapters	Bloomfield SA. "The risks to bone health during missions to the Moon and Mars." in "Bone loss during spaceflight : etiology, countermeasures, and implications for bone health on Earth." Ed. P.R. Cavanagh, A.J. Rice. Cleveland : Cleveland Clinic Press, c2007. p. 229-237., Sep-2007